

[14X030] Introduction to Computational Finance

Exercise #13

May 26, 2026

General instructions

Each student is expected to upload on Moodle a **zip** file named as *firstname_lastname*, containing the followings:

- A report in **pdf** format to answer exercise questions.
- The **code** used to generate the results of the report.
- If preferred, submit a unique **Jupyter notebook** with both code and report explanations.

Deeper instructions

- **Report:**
 - Answers to the TP questions. Figures and numerical results are necessary but **strictly not** sufficient: provide your analysis/comments as well.
 - Example: if your results surprise you, explain why and explain what the expected result was.
 - **Pay attention** to the presentation: axis labels, legend, etc.
- **Code:**
 - Implementation questions + code used to produce the results and figures in the report.
 - Programming language of your choice.
 - Do not use ChatGPT, or any other LLM/AI-assisted tools while writing down your code. Try by yourself, otherwise it will be counted as failed.
 - I **strongly recommend Python/Julia**: I won't be able to assist you in the same way if you use a different language.

Deadline

Any questions to: [Lorenzo Bini](#).

Upload on Moodle due by : **June 01, 2026 at 11:59 pm**

Futures valuation

We consider a non-dividend-paying stock X that is valued at $S_0 = 100$ today, and the risk-free rate per year is $r = 0.1$. Furthermore, we assume the price S_t of X to evolve along two scenarios as follows:

t [month]	price scenario 1	price scenario 2
0	100	100
2	105	99
4	104	101
6	108	98
8	113	97
10	109	99
12	112	97

We consider a future with maturity $T = 12$ months.

- Compute the future value F_t at each date for both scenarios as time passes by.
- Plot the evolution of S_t and F_t on the same graph.
- What can you say about the way F_t converges towards S_T ?

Mark-to-Market P/L of a Futures Position

In this exercise, you will track the monthly profit and loss of a long futures position, marked to market at each observation date, and see how variation margin accumulates over time.

1. **Futures Prices:** You already computed $F_t = S_t \exp(r(T - t))$ at each monthly date t for both scenarios in Exercise #1.
2. **Mark-to-Market P/L:**
 - Assume you enter a long futures contract at $t_0 = 0$ at price F_0 .
 - At each subsequent date t_i , your variation margin (PnL) is

$$\Delta P_i = F_{t_i} - F_{t_{i-1}}.$$

- Compute ΔP_i for $i = 1, 2, \dots, 6$ (every 2 months), and the cumulative P/L $\text{CumPL}_i = \sum_{j=1}^i \Delta P_j$.

3. **Plotting & Discussion:**

- On one chart, plot the two scenarios of cumulative P/L versus t .
- On a second chart, overlay ΔP_i (bar chart) for each scenario.
- Comment on the variability of monthly variation margins under each scenario.